

Color version of the adaptive double-phase ROF model: report

Wojciech Górny* Michał Łasica† Alexandros Matsoukas‡

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1 Vectorial formulation of ROF and dpROF for RGB images

We give here a short description of the ROF and dpROF models in the vector-valued RGB setting. The main point is to describe the vectorial extension of the two models and to show how the Chambolle–Pock algorithm can be adapted to this setting. Let

$$u = (R, G, B) : \Omega \rightarrow \mathbb{R}^3$$

be an RGB image. At each pixel, its spatial derivative is represented by the Jacobian matrix

$$\nabla u_{i,j} = \begin{pmatrix} D_x R_{i,j} & D_y R_{i,j} \\ D_x G_{i,j} & D_y G_{i,j} \\ D_x B_{i,j} & D_y B_{i,j} \end{pmatrix} \in \mathbb{R}^{3 \times 2}.$$

The Frobenius norm of this matrix is

$$\|\nabla u_{i,j}\|_F = \sqrt{|\nabla R_{i,j}|^2 + |\nabla G_{i,j}|^2 + |\nabla B_{i,j}|^2}.$$

Hence, the spatial information from the three color channels contributes jointly to a common gradient magnitude. In contrast to applying a scalar regularizer independently to each color channel, the vectorial formulation uses this common quantity in the regularization term. In particular, an edge that is present in one or more channels has a direct influence on the regularization of the RGB image, through the common Frobenius gradient magnitude.

The vectorial ROF model is given by

$$E_{\text{ROF}}(u) = \sum_{i,j} \|\nabla u_{i,j}\|_F + \frac{1}{2\lambda} \|u - f\|_2^2.$$

For an RGB image, the fidelity term is understood as

$$\|u - f\|_2^2 = \sum_{i,j} \|u_{i,j} - f_{i,j}\|_{\mathbb{R}^3}^2.$$

The first term promotes spatial regularity through the Frobenius total variation, while the fidelity term keeps the reconstruction close to the observed noisy image. The parameter λ controls the balance between the fidelity and the regularizer.

The vectorial dpROF model retains the Frobenius total variation structure and introduces an additional spatially varying weighted quadratic term:

$$E_{\text{dpROF}}(u) = \sum_{i,j} \left(\|\nabla u_{i,j}\|_F + \frac{w_{i,j}}{2} \|\nabla u_{i,j}\|_F^2 \right) + \frac{1}{2\lambda} \|u - f\|_2^2.$$

The adaptive weight $w_{i,j}$ is constructed from a smoothed vectorial ROF reconstruction. More precisely, after smoothing u_{ROF} with the mollifying kernel ρ_r , we define

$$w_{i,j} = W \left(\|\nabla(\rho_r * u_{\text{ROF}})_{i,j}\|_F \right),$$

where W is the same cutoff-type function as in the scalar model i.e., the function W is given by

$$W_1(x) = \max \left(0, a - b \max \left(x, \frac{a}{2b} \right) \right).$$

*Faculty of Mathematics, Universität Wien, Oskar-Morgerstern-Platz 1, 1090 Vienna, Austria; Faculty of Mathematics, Informatics and Mechanics, University of Warsaw, Banacha 2, 02-097 Warsaw, Poland; wojciech.gorny@univie.ac.at

†Institute of Mathematics of the Polish Academy of Sciences, Śniadeckich 8, 00-656 Warsaw, Poland; mlasica@impan.pl

‡Department of Mathematics, School of Applied Mathematical and Physical Sciences, National Technical University of Athens, Zografou Campus, 157 80 Athens, Greece; alexmatsoukas@mail.ntua.gr

The use of the Frobenius norm allows the weight to depend on the gradient information from all three color channels. In particular, if the common gradient magnitude is large, the weight becomes smaller and the linear-growth part of the regularizer is dominant near edges. In smoother regions, where the gradient magnitude is smaller, the weight becomes larger and the quadratic part has a stronger effect. This also allows edges that are mainly present in the chromatic information to affect the adaptive weight.

The Frobenius norm is invariant under orthogonal transformations and depends only on the magnitude of the matrix entries. Moreover, this radial structure is also important computationally, as it allows us to extend the Chambolle–Pock algorithm directly from the scalar to the vectorial case. The form of the iterations remains the same, while the Euclidean norm appearing in the closed-form dual resolvents of the ROF and dpROF models is replaced by the Frobenius norm.

Resolvents in the Chambolle–Pock algorithm

Both models can be written in the form

$$E(u) = F(\nabla u) + G(u),$$

where F is the regularization term and

$$G(u) = \frac{1}{2\lambda} \|u - f\|_2^2$$

is the fidelity term. In the Chambolle–Pock algorithm, the dual update uses the resolvent of ∂F^* , while the primal update uses the proximal operator of G . Therefore, below we compute the dual resolvents associated with the ROF and dpROF regularizers. For the fidelity term, the proximal operator is immediate. For any vector-valued image v ,

$$\text{prox}_{\tau G}(v) = \frac{v + (\tau/\lambda)f}{1 + \tau/\lambda}.$$

This formula is applied componentwise to the RGB image and is the same as in the scalar case.

Dual resolvent for the vectorial ROF model

For the regularization term, define

$$\varphi(Z) = \|Z\|_F, \quad Z \in \mathbb{R}^{3 \times 2}.$$

Its Fenchel conjugate is

$$\varphi^*(P) = \iota_{\{\|P\|_F \leq 1\}}(P),$$

where ι_C is the indicator function of a set C ,

$$\iota_C(P) = \begin{cases} 0, & P \in C, \\ +\infty, & P \notin C. \end{cases}$$

Hence, the dual resolvent appearing in the Chambolle–Pock update is simply the projection onto the Frobenius unit ball. If $\tilde{P}$ denotes the dual variable before the proximal step, then

$$P = \frac{\tilde{P}}{\max\{1, \|\tilde{P}\|_F\}}.$$

This gives the dual projection used in the vectorial ROF model, in the same way as in the scalar case. The Euclidean norm of the scalar dual vector is replaced by the Frobenius norm of the 3×2 dual matrix.

Dual resolvent for the vectorial dpROF model

At a fixed pixel, define

$$\varphi_{\text{dp}}(Z) = \|Z\|_F + \frac{w}{2} \|Z\|_F^2, \quad Z \in \mathbb{R}^{3 \times 2}.$$

We now give a short description of the derivation of the corresponding dual resolvent in the vector-valued case. For $w = 0$,

$$\varphi^*(P) = \iota_{\{\|P\|_F \leq 1\}}(P),$$

so the same projection on the Frobenius ball as in vectorial ROF is obtained. For $w > 0$, the Fenchel conjugate is

$$\varphi_{\text{dp}}^*(P) = \sup_Z \left\{ \langle P, Z \rangle_F - \|Z\|_F - \frac{w}{2} \|Z\|_F^2 \right\}.$$

By the Frobenius Cauchy–Schwarz inequality,

$$\langle P, Z \rangle_F \leq \|P\|_F \|Z\|_F,$$

the maximizer is parallel to the dual variable. Hence, the computation of the vectorial dual resolvent reduces to the same radial one-dimensional problem as in the scalar case, with the Euclidean norm replaced by the Frobenius norm. Therefore, the closed-form resolvent is given by

$$P_{i,j} = \begin{cases} \frac{\tilde{P}_{i,j}}{\max\{1, \|\tilde{P}_{i,j}\|_F\}}, & w_{i,j} = 0, \\ \min\left\{1, \frac{w_{i,j}\|\tilde{P}_{i,j}\|_F + \sigma}{w_{i,j}\|\tilde{P}_{i,j}\|_F + \sigma\|\tilde{P}_{i,j}\|_F}\right\} \tilde{P}_{i,j}, & w_{i,j} > 0. \end{cases}$$

Thus, the vectorial dpROF resolvent has the same radial form as in the scalar model. The same scalar factor is applied to all six derivative components of the 3×2 dual matrix, which couples the three color channels in the dual update.

2 Optimisation of parameters for vectorial RGB ROF

Table 1: Optimal values of λ for the vectorial ROF model on the training set according to LPIPS, SSIM, and PSNR.

σ^2	LPIPS λ	SSIM λ	PSNR λ
0.001	0.8σ	1.0σ	0.9σ
0.002	0.8σ	1.0σ	0.9σ
0.004	1.0σ	1.1σ	1.0σ
0.007	1.0σ	1.1σ	1.0σ
0.010	1.0σ	1.1σ	1.1σ
0.020	1.1σ	1.2σ	1.1σ
0.040	1.2σ	1.2σ	1.1σ
0.070	1.2σ	1.2σ	1.1σ

Based on the parameter-search results, we use a simplified parameter prescription with $\lambda = 1.0\sigma$, 1.1σ , and 1.2σ for small, medium, and large noise levels, respectively.

Table 2: Selected parameter values for the vectorial ROF model.

Noise level	λ
Small	1.0σ
Medium	1.1σ
Large	1.2σ

3 Optimisation of parameters for vectorial RGB dpROF

In this subsection, we present the parameter selection for the vectorial dpROF model using the BSDS500 dataset. The aim is to study how the model parameters change with the noise level and to obtain a simple rule for the final choice of parameters. We first use the training set to identify suitable parameter values and then use the validation set to refine the final parameter selection. We first performed a parameter search on the training set for each noise level. The parameters were tested over predefined grids, and the values giving the best LPIPS, SSIM, and PSNR were recorded separately. We then repeated the same procedure on the validation set, using a smaller search around the values obtained from the training set. This allowed us to compare the parameter choices on the two datasets and to identify common trends across the different noise levels.

Table 3: Optimal parameters of the vectorial adaptive double-phase ROF model on the training set.

σ^2	LPIPS			SSIM			PSNR		
	λ	a	b/a	λ	a	b/a	λ	a	b/a
0.001	0.5σ	90	100	0.8σ	90	110	0.8σ	90	110
0.002	0.6σ	100	90	0.8σ	100	80	0.8σ	100	80
0.004	0.6σ	100	60	0.8σ	110	60	0.8σ	110	60
0.007	0.7σ	90	50	0.9σ	110	60	0.9σ	100	50
0.010	0.7σ	100	50	0.9σ	90	50	0.9σ	100	50
0.020	0.8σ	110	50	0.9σ	80	50	0.9σ	120	50
0.040	0.9σ	200	50	0.9σ	110	50	1.0σ	80	50
0.070	1.0σ	200	50	0.9σ	90	50	0.9σ	110	50

Table 4: Optimal parameters of the vectorial adaptive double-phase ROF model on the validation set.

σ^2	LPIPS			SSIM			PSNR		
	λ	a	b/a	λ	a	b/a	λ	a	b/a
0.001	0.5σ	100	110	0.8σ	90	110	0.7σ	100	100
0.002	0.5σ	100	80	0.8σ	100	90	0.8σ	100	90
0.004	0.5σ	100	60	0.8σ	90	60	0.8σ	90	60
0.007	0.7σ	110	60	0.8σ	100	50	0.9σ	110	60
0.010	0.7σ	80	50	0.8σ	90	50	0.9σ	80	50
0.020	0.8σ	80	50	0.9σ	80	50	0.9σ	90	50
0.040	0.9σ	80	50	0.9σ	80	50	1.0σ	80	50
0.070	0.9σ	200	50	0.9σ	90	50	0.9σ	90	50

Table 5: Grouped parameter prescription obtained from the training set.

Noise level	λ	a	b/a
Small	0.8σ	100	80
Medium	0.9σ	100	50
High	0.9σ	110	50

To obtain a simpler parameter rule, we grouped the noise levels into three classes: small, medium, and high noise. For each noise level, we selected a representative value of λ from the LPIPS, SSIM, and PSNR optimal parameter values, and we chose a and b/a close to the mean and median of the corresponding optimal parameter values. We then grouped the resulting representative parameters across the three noise ranges.

We applied the same procedure to the validation set optimal parameter values. Again, the noise levels were grouped into small, medium, and high noise, and representative values were selected from the LPIPS, SSIM, and PSNR optimal parameter values.

Table 6: Grouped parameter prescription obtained from the validation set.

Noise level	λ	a	b/a
Small	0.8σ	100	90
Medium	0.8σ	90–100	50–60
High	0.9σ	80–90	50

The grouped training and validation results show similar trends. In particular, λ increases with the noise level, while b/a decreases and becomes stable around 50 for medium and high noise. The parameter a varies more, but most optimal values remain close to 100. Based on these common trends, we use the following simplified parameter prescription for the vectorial dpROF model.

Table 7: Final simplified parameter prescription obtained from the training and validation sets.

Noise level	λ_{ROF}	λ_{dpROF}	a	b/a
Small	1.0σ	0.8σ	100	80
Medium	1.1σ	0.9σ	100	50
High	1.2σ	0.9σ	100	50

3.1 Sensitivity with respect to λ .

To further examine the sensitivity of the vectorial dpROF model with respect to λ , we consider three representative noise levels corresponding to small, medium, and high noise. For each value of λ , we report the best LPIPS, SSIM, and PSNR values obtained over the corresponding grid of the remaining parameters. The results show that LPIPS generally favors smaller values of λ at low and medium noise levels, while SSIM and PSNR tend to prefer slightly larger values. As the noise level increases, the preferred values of λ become closer across the three metrics.

Table 8: Sensitivity with respect to λ for noise variance $\sigma^2 = 0.004$.

λ/σ	LPIPS $\downarrow$	SSIM $\uparrow$	PSNR $\uparrow$
0.5	0.123010	0.844706	29.8931
0.6	0.121523	0.853924	30.2090
0.7	0.121892	0.858189	30.3862
0.8	0.124450	0.860179	30.4642
0.9	0.128923	0.860105	30.4694

Table 9: Sensitivity with respect to λ for noise variance $\sigma^2 = 0.01$.

λ/σ	LPIPS $\downarrow$	SSIM $\uparrow$	PSNR $\uparrow$
0.6	0.207142	0.779309	27.8746
0.7	0.205562	0.788589	27.9891
0.8	0.206840	0.793473	28.1244
0.9	0.208661	0.793707	28.1800
1.0	0.212506	0.792088	28.1469

Table 10: Sensitivity with respect to λ for noise variance $\sigma^2 = 0.04$.

λ/σ	LPIPS $\downarrow$	SSIM $\uparrow$	PSNR $\uparrow$
0.7	0.370854	0.658509	24.5933
0.8	0.364464	0.665332	24.6558
0.9	0.361473	0.667786	24.6888
1.0	0.361560	0.666478	24.6911
1.1	0.364290	0.662201	24.6290
1.2	0.367846	0.656302	24.5316

The selected simplified values of λ provide a good compromise between the three metrics. For $\sigma^2 = 0.004$, using $\lambda = 0.8\sigma$ gives the optimal SSIM, while the PSNR is only 0.0052 dB below its optimum. For $\sigma^2 = 0.01$, $\lambda = 0.9\sigma$ gives the optimal SSIM and PSNR, with only a small increase in LPIPS compared with its minimum. Finally, for $\sigma^2 = 0.04$, $\lambda = 0.9\sigma$ gives the optimal LPIPS and SSIM, while the PSNR differs from its optimum by only 0.0023 dB. These results support the use of a single simplified parameter prescription derived from the BSDS500 training set for the vectorial dpROF model.

3.2 Sensitivity with respect to a and b/a

To study the sensitivity of the vectorial dpROF model with respect to the parameters a and b/a , we consider three representative noise levels. For each metric, λ is fixed to the value selected during the parameter search, while a and b/a are varied around the corresponding selected values. The results indicate that the model is relatively stable with respect to a , with a slightly stronger dependence on b/a .

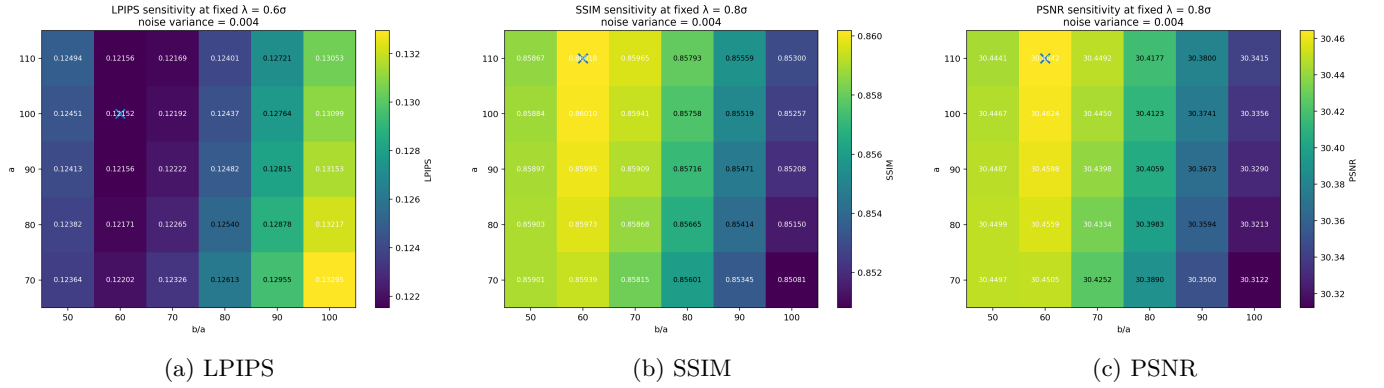


Figure 1: Sensitivity with respect to a and b/a for noise variance $\sigma^2 = 0.004$.

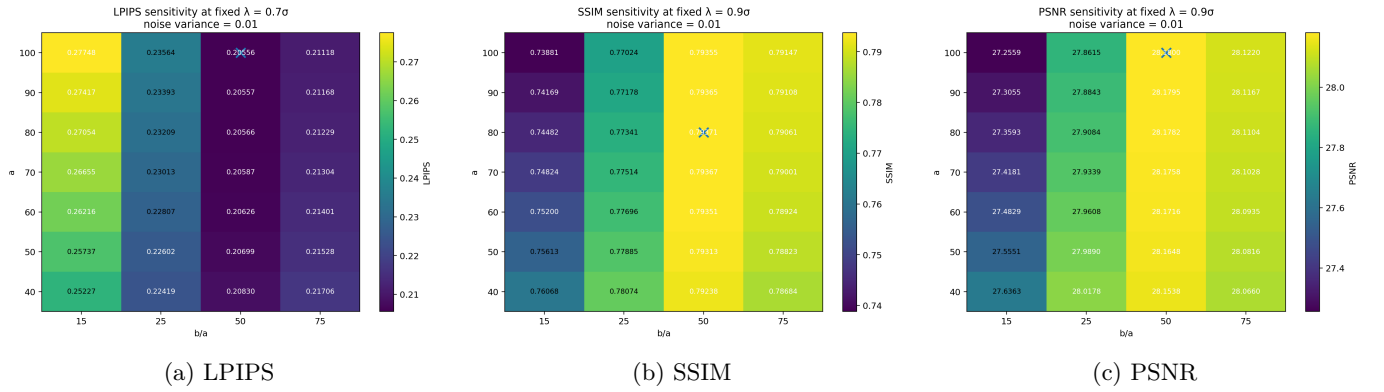


Figure 2: Sensitivity with respect to a and b/a for noise variance $\sigma^2 = 0.01$.

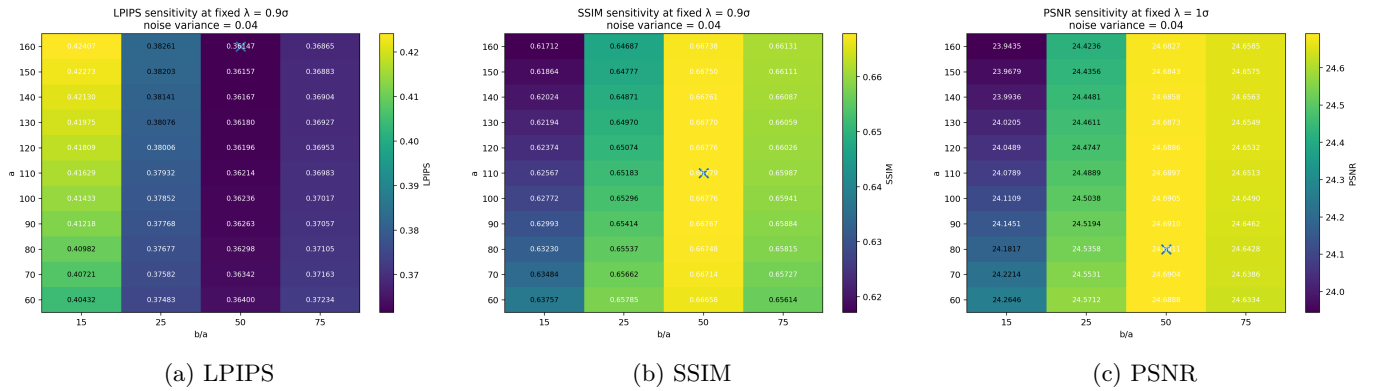


Figure 3: Sensitivity with respect to a and b/a for noise variance $\sigma^2 = 0.04$.

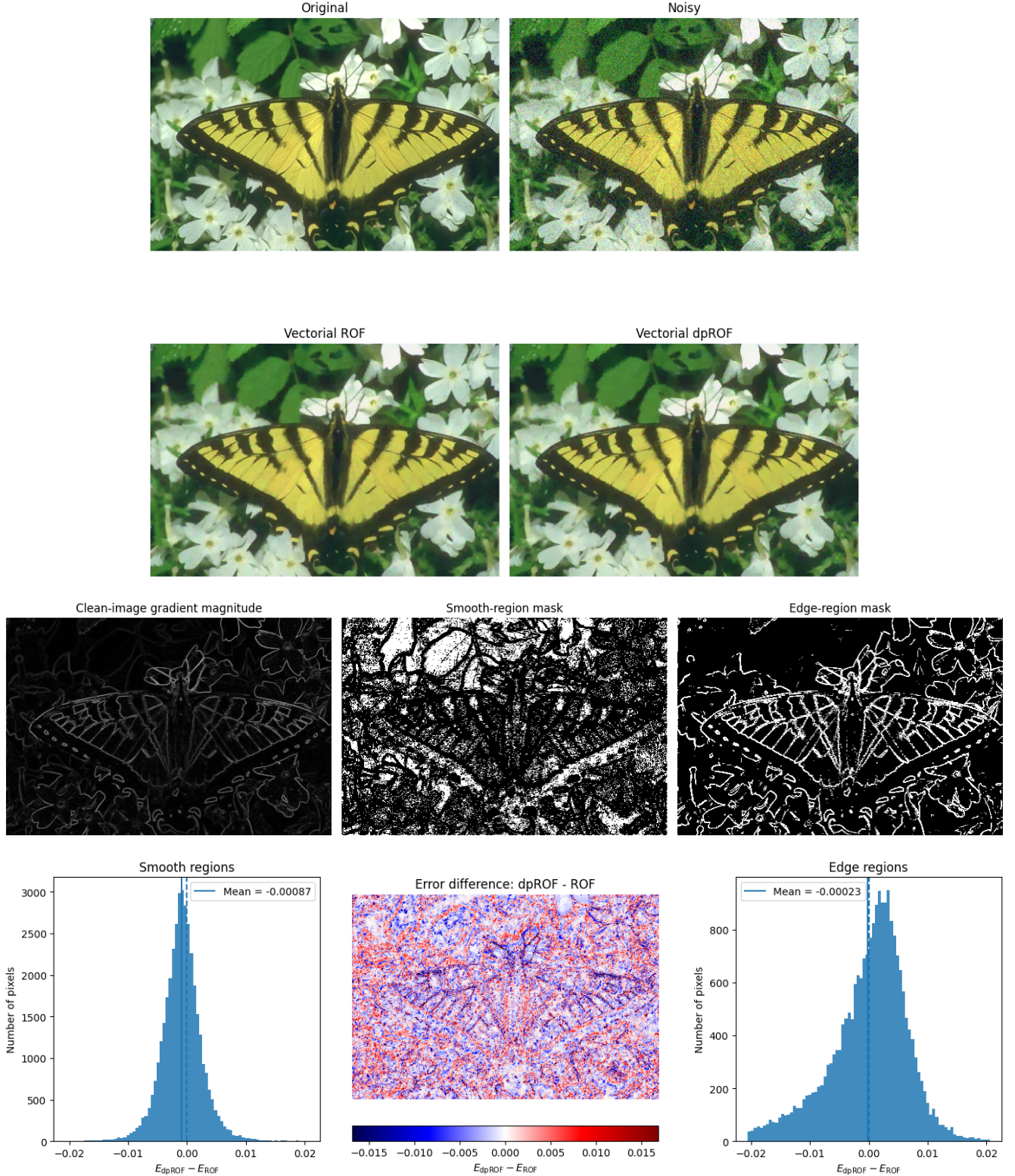
Illustrative diagnostics for the vectorial dpROF model

We provide three local examples comparing vectorial ROF and dpROF, with $\sigma^2 = 0.01$ and parameters obtained from the BSDS500 training set, for illustrative purposes only. Smooth and edge regions are defined from the gradient magnitude of the clean image. The error-difference map is

$$D(x) = E_{\text{dpROF}}(x) - E_{\text{ROF}}(x),$$

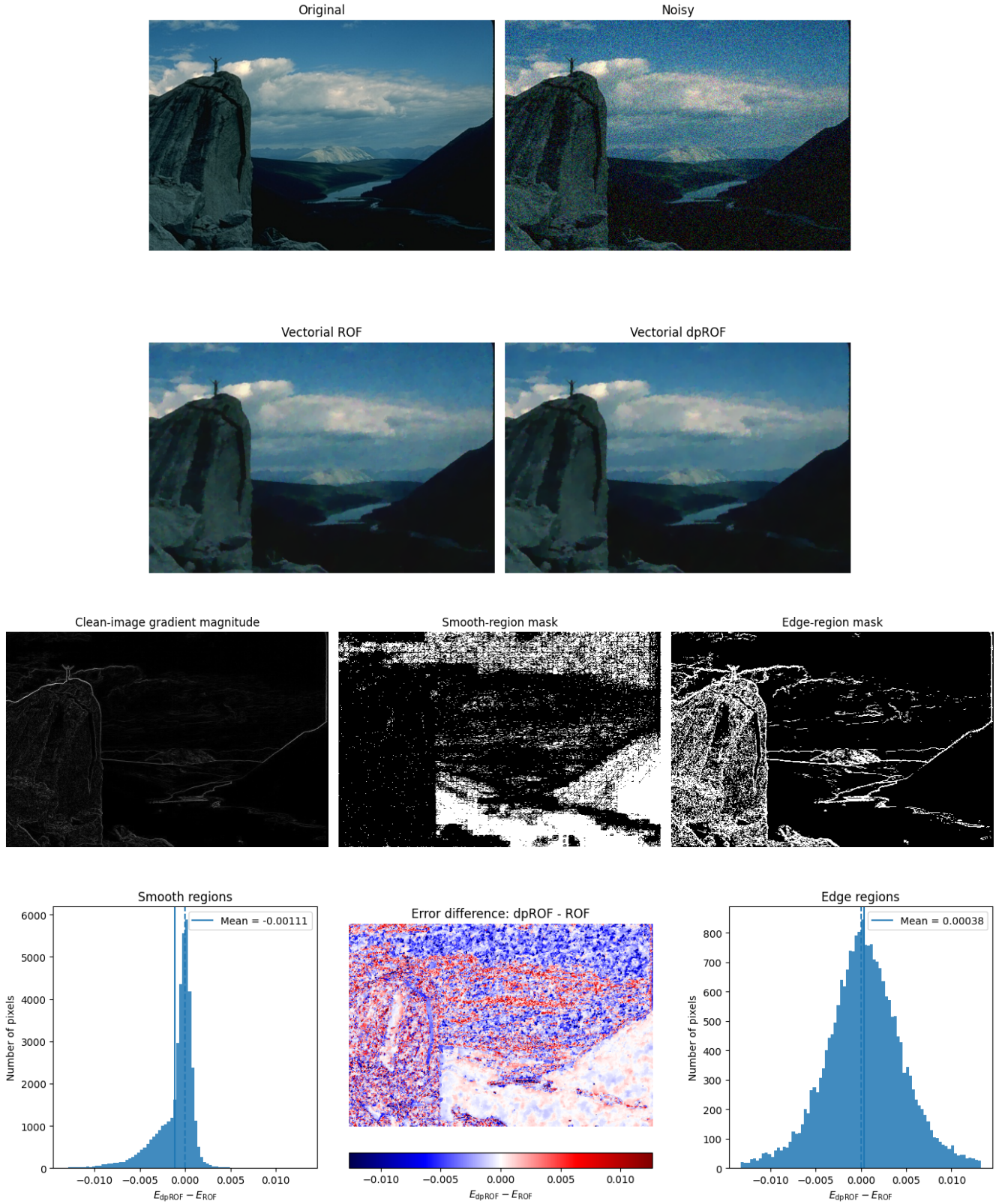
so negative values indicate a smaller local reconstruction error for dpROF.

Example 1: Butterfly



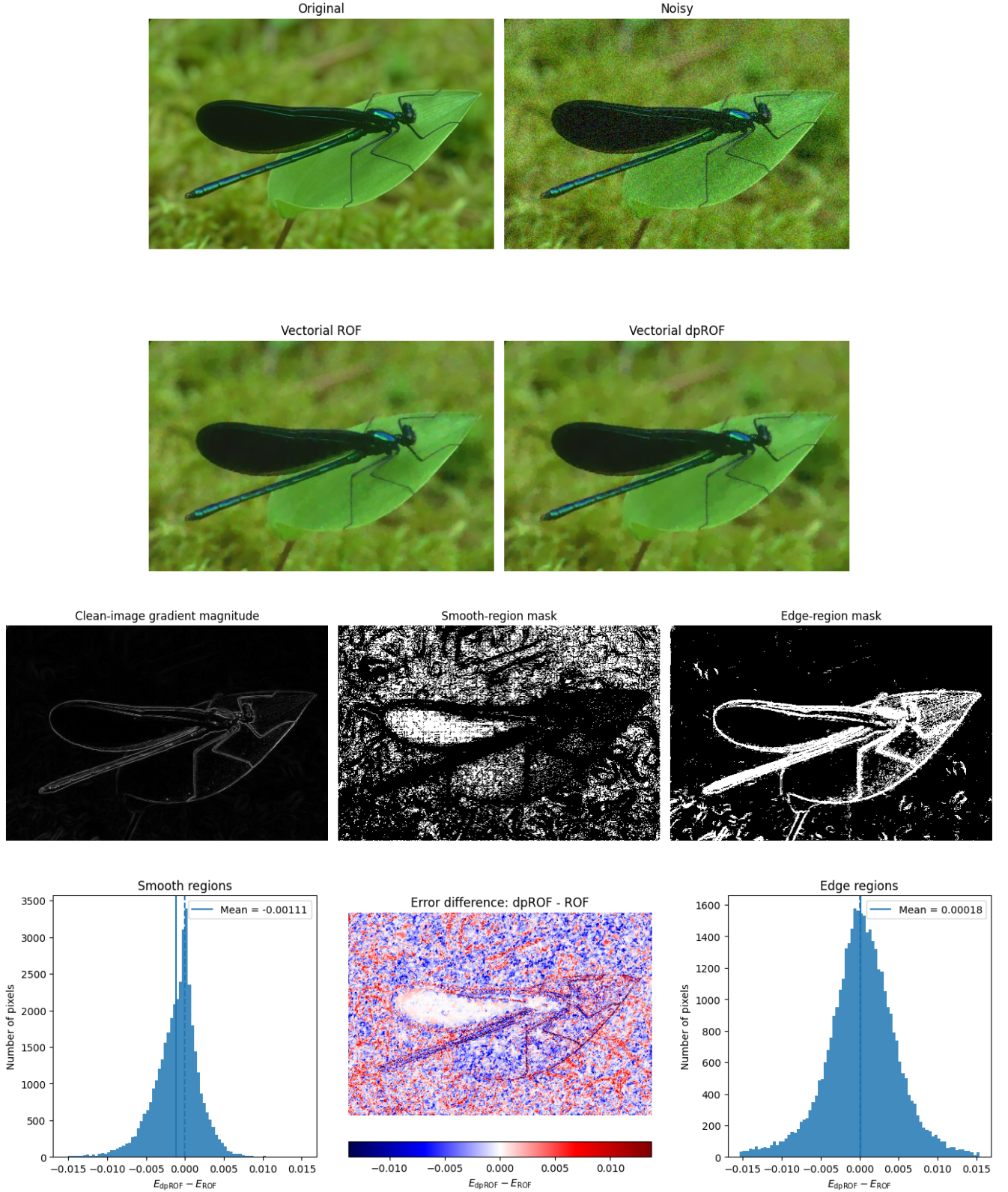
In smooth regions, dpROF gives a smaller reconstruction error for 62.85% of the pixels. In edge regions, ROF is smaller for 56.17% of the pixels, while the mean error difference remains close to zero. This indicates a clearer dpROF advantage in smooth areas, while the two methods remain comparable near strong edges.

Example 2: Mountain



In smooth regions, dpROF gives a smaller reconstruction error for 63.86% of the pixels. In edge regions, the two methods are again close, with ROF smaller for 53.72% of the pixels. The main difference is therefore observed in locally homogeneous regions.

Example 3: Dragonfly



In smooth regions, dpROF gives a smaller reconstruction error for 62.54% of the pixels. In edge regions, the two methods are almost balanced, with dpROF smaller for 48.70% and ROF smaller for 51.30% of the pixels. This again suggests that the main local improvement of dpROF is concentrated in smooth regions, while edge-region errors remain similar.

4 Comparison of different models in the color case

In this section we compare the test results between several models for color images: the classical ROF model (ROF), the adaptive double-phase ROF model (dpROF), and the non-local means model (NL-means). As in the case of grayscale images, we first describe the choice of all parameters. The mollification radius is always set to $r = 2$; the

prescribed accuracy levels ε_{ROF} (i.e., the algorithm stops if the L^2 distance between two iterations in a row is lower than this value) and $\varepsilon_{\text{dpROF}}$ for the classical and double-phase ROF models are set to 10^{-4} , with the exception that the initial ROF for the calculation of the weight in dpROF has accuracy 10^{-2} . Below, we describe how we chose the values of λ_{ROF} in the Rudin–Osher–Fatemi model, λ to be used in the double-phase ROF model, and the weight parameters a and b .

Table 11: Choice of parameters in the vectorial adaptive double-phase ROF model

Noise level	Value of λ_{ROF}	Value of λ	Value of a	Value of b/a
Small ($\sigma^2 \leq 0.005$)	$1.0 \cdot \sigma$	$0.8 \cdot \sigma$	100	85
Medium ($0.005 < \sigma^2 \leq 0.015$)		$0.9 \cdot \sigma$		50
Large ($\sigma^2 > 0.015$)	$1.2 \cdot \sigma$			50

This parameter choice was obtained in a very simple way: first we observe that the variability in a was very small, thus we pick $a = 100$ for all noise levels to simplify the model (it is the median value over the optimal parameters returned over “train” and “val” across all noise levels). for each noise level, we took the median value of the optimal parameters returned over the “train” and “val” datasets (rounded up to the first decimal value). In a similar fashion, we picked the parameter h in the non-local means model (equal to 0.4 for small and medium noise levels and to 0.45 for large noise levels). We additionally consider performance of channel-wise dpROF model applied to YUV and RGB color schemes (using the parameters from the grayscale dpROF model); these we denote YUV-dpROF and RGB-dpROF respectively. The YUV-dpROF model uses a single weight computed from the Y (luminance) channel of the datum, while RGB-dpROF uses a different weight computed as in the grayscale case separately for each channel.

We now present the comparison of the three vectorial models, together with the two channel-wise reconstructions, done by performing a single run of all the models over the “test” part of the BSDS500 dataset. Let us note that the channel-wise versions, while taking approximately 3x longer (due to similar algorithms applied separately on the three channels), overall perform worse in all metrics than the standard vectorial dpROF model across 7/8 noise levels (for YUV-dpROF) and 8/8 noise levels (for RGB-dpROF).

Table 12: Comparison of denoising methods: $\sigma^2 = 0.001$

Method	Metrics			
	Avg. time	Avg. SSIM $\uparrow$	Avg. LPIPS $\downarrow$	Avg. PSNR $\uparrow$
ROF	0.77s	0.9289	0.0586	33.90
dpROF	0.25s + 1.42s	0.9312	0.0543	34.11
NL-means	0.25s	0.9236	0.0653	33.29
YUV-dpROF	4.29s	0.9283	0.0712	34.08
RGB-dpROF	4.37s	0.9187	0.0640	33.37

Table 13: Comparison of denoising methods: $\sigma^2 = 0.002$

Method	Metrics			
	Avg. time	Avg. SSIM $\uparrow$	Avg. LPIPS $\downarrow$	Avg. PSNR $\uparrow$
ROF	0.83s	0.8970	0.0876	31.99
dpROF	0.27s + 1.55s	0.9008	0.0796	32.15
NL-means	0.26s	0.8928	0.0916	31.21
YUV-dpROF	4.77s	0.8986	0.1012	32.22
RGB-dpROF	4.68s	0.8839	0.0937	31.45

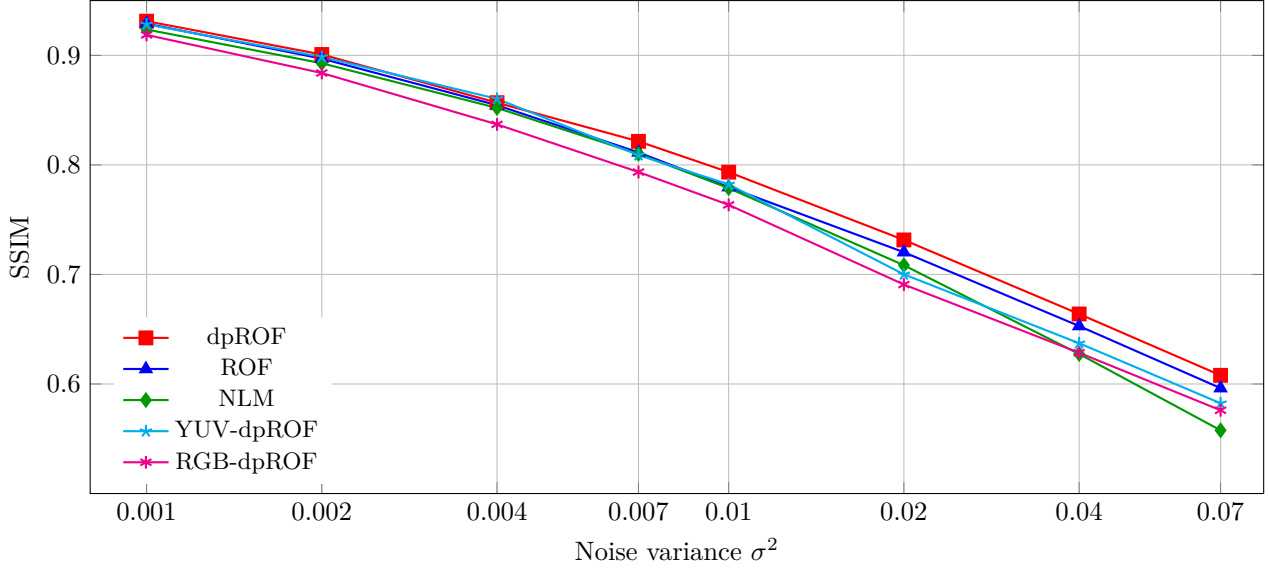
Table 14: Comparison of denoising methods: $\sigma^2 = 0.004$

Method	Metrics			
	Avg. time	Avg. SSIM $\uparrow$	Avg. LPIPS $\downarrow$	Avg. PSNR $\uparrow$
ROF	0.93s	0.8545	0.1301	30.16
dpROF	0.29s + 1.65s	0.8568	0.1217	30.21
NL-means	0.28s	0.8520	0.1273	29.33
YUV-dpROF	5.57s	0.8603	0.1413	30.44
RGB-dpROF	5.33s	0.8370	0.1389	29.59

Table 15: Comparison of denoising methods: $\sigma^2 = 0.007$

Method	Metrics			
	Avg. time	Avg. SSIM $\uparrow$	Avg. LPIPS $\downarrow$	Avg. PSNR $\uparrow$
ROF	1.01s	0.8114	0.1768	28.73
dpROF	0.32s + 1.92s	0.8215	0.1744	28.84
NL-means	0.29s	0.8101	0.1635	27.96
YUV-dpROF	6.35s	0.8089	0.2017	28.74
RGB-dpROF	5.87s	0.7935	0.1983	28.16

Figure 4: Comparison of ROF, dpROF, NLM, YUV- and RGB-dpROF models across different noise levels (SSIM)

Table 16: Comparison of denoising methods: $\sigma^2 = 0.01$

Method	Metrics			
	Avg. time	Avg. SSIM $\uparrow$	Avg. LPIPS $\downarrow$	Avg. PSNR $\uparrow$
ROF	1.08s	0.7797	0.2124	27.83
dpROF	0.35s + 2.06s	0.7934	0.2057	27.95
NL-means	0.30s	0.7788	0.1900	27.13
YUV-dpROF	6.91s	0.7822	0.2307	27.91
RGB-dpROF	6.38s	0.7635	0.2315	27.30

Table 17: Comparison of denoising methods: $\sigma^2 = 0.02$

Method	Metrics			
	Avg. time	Avg. SSIM $\uparrow$	Avg. LPIPS $\downarrow$	Avg. PSNR $\uparrow$
ROF	1.63s	0.7203	0.2895	26.06
dpROF	0.43s + 2.26s	0.7315	0.2829	26.24
NL-means	0.32s	0.7084	0.2631	25.84
YUV-dpROF	8.15s	0.7001	0.3231	25.79
RGB-dpROF	7.65s	0.6908	0.3267	25.59

Table 18: Comparison of denoising methods: $\sigma^2 = 0.04$

Method	Metrics			
	Avg. time	Avg. SSIM $\uparrow$	Avg. LPIPS $\downarrow$	Avg. PSNR $\uparrow$
ROF	1.99s	0.6529	0.3673	24.28
dpROF	0.51s + 2.71s	0.6639	0.3647	24.44
NL-means	0.33s	0.6274	0.3406	24.17
YUV-dpROF	9.72s	0.6372	0.3929	24.09
RGB-dpROF	9.11s	0.6285	0.3978	23.98

Table 19: Comparison of denoising methods: $\sigma^2 = 0.07$

Method	Metrics			
	Avg. time	Avg. SSIM $\uparrow$	Avg. LPIPS $\downarrow$	Avg. PSNR $\uparrow$
ROF	2.40s	0.5962	0.4331	22.63
dpROF	0.61s + 3.15s	0.6079	0.4311	22.84
NL-means	0.34s	0.5577	0.4221	22.58
YUV-dpROF	11.33s	0.5824	0.4560	22.46
RGB-dpROF	10.61s	0.5760	0.4608	22.47

Figure 5: Comparison of ROF, dpROF, NLM, YUV- and RGB-dpROF models across different noise levels (LPIPS)

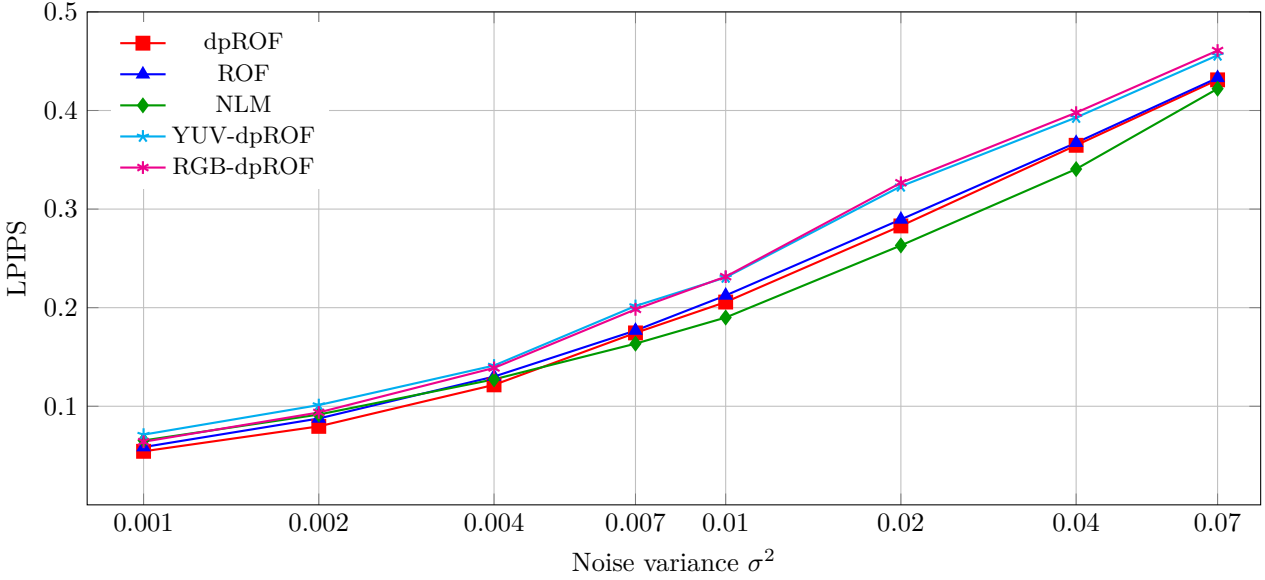
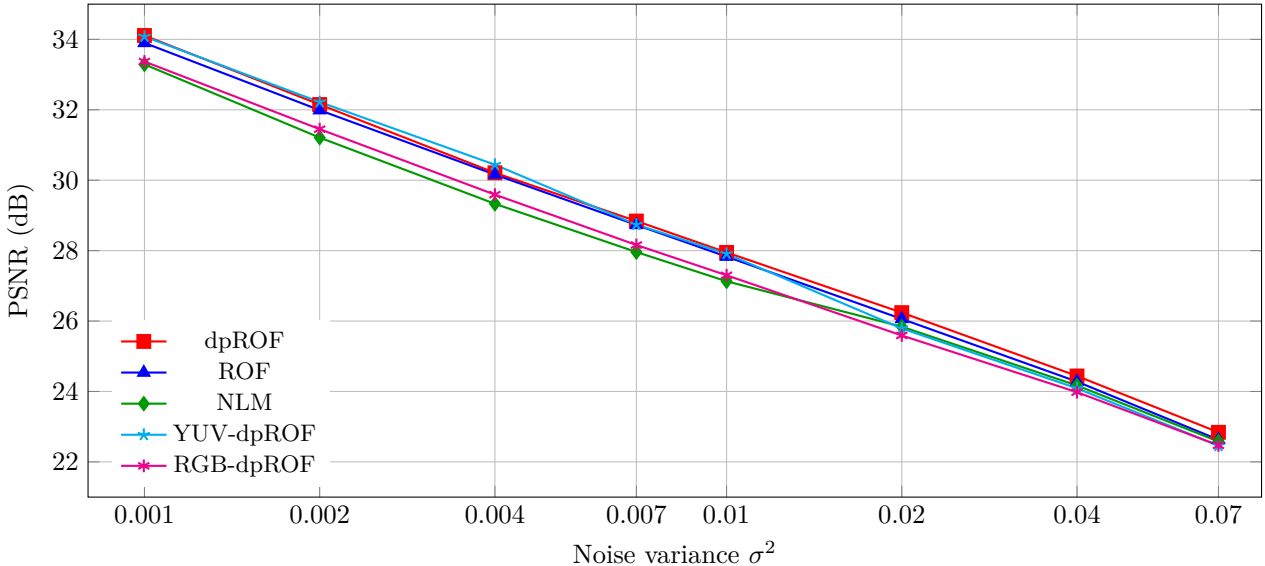


Figure 6: Comparison of ROF, dpROF, NLM, YUV- and RGB-dpROF models across different noise levels (PSNR)



5 Cross-dataset performance

Similarly to the grayscale case (presented in the ‘comparison’ appendix), to evaluate the generalization of the proposed parameter selection, we test the parameters obtained from BSDS500 on four unseen datasets: Kodak24, Urban100, EuroSAT and Set14 (in two different resolutions); see [2], [4], [1] and [3] respectively. Again, the parameters are transferred directly from BSDS500 without any dataset-specific retuning. The experiments are performed for all considered noise levels, and the resulting SSIM, PSNR, and LPIPS values are reported below. In particular, the transferred prescription is given by Table 11, depending on the considered noise regime.

Table 20: Cross-dataset performance of vectorial dpROF using the parameter prescription derived from BSDS500. No dataset-specific parameter retuning was performed on Kodak24, Urban100, EuroSAT or Set14.

Dataset	Metric	Noise variance σ^2							
		0.001	0.002	0.004	0.007	0.01	0.02	0.04	0.07
BSDS500	SSIM $\uparrow$	0.9312	0.9008	0.8568	0.8215	0.7934	0.7315	0.6639	0.6079
	LPIPS $\downarrow$	0.0543	0.0796	0.1217	0.1744	0.2057	0.2829	0.3647	0.4311
	PSNR $\uparrow$	34.11	32.15	30.21	28.84	27.95	26.24	24.44	22.84
Kodak24	SSIM $\uparrow$	0.9207	0.8907	0.8472	0.8167	0.7911	0.7385	0.6799	0.6339
	LPIPS $\downarrow$	0.0801	0.1081	0.1540	0.2122	0.2446	0.3207	0.4001	0.4626
	PSNR $\uparrow$	34.61	32.76	30.88	29.64	28.78	27.16	25.39	23.82
Urban100	SSIM $\uparrow$	0.9426	0.9165	0.8791	0.8442	0.8181	0.7563	0.6838	0.6148
	LPIPS $\downarrow$	0.0261	0.0416	0.0689	0.1049	0.1296	0.1947	0.2750	0.3518
	PSNR $\uparrow$	33.26	31.13	29.03	27.35	26.38	24.48	22.56	20.93
EuroSAT	SSIM $\uparrow$	0.9313	0.9022	0.8555	0.8407	0.8145	0.7737	0.7222	0.6837
	LPIPS $\downarrow$	0.0438	0.0565	0.0803	0.1134	0.1305	0.1783	0.2282	0.2693
	PSNR $\uparrow$	36.95	35.11	32.99	32.57	31.59	30.26	28.23	26.43
Set14 (low)	SSIM $\uparrow$	0.9220	0.8912	0.8485	0.8139	0.7871	0.7281	0.6644	0.6102
	LPIPS $\downarrow$	0.0610	0.0825	0.1155	0.1627	0.1890	0.2545	0.3269	0.3891
	PSNR $\uparrow$	33.98	32.05	30.11	28.76	27.85	26.07	24.19	22.52
Set14 (high)	SSIM $\uparrow$	0.9448	0.9196	0.8800	0.8784	0.8570	0.8281	0.7870	0.7575
	LPIPS $\downarrow$	0.0645	0.1019	0.1715	0.1726	0.2143	0.2553	0.3222	0.3607
	PSNR $\uparrow$	37.98	35.84	33.55	32.52	31.36	29.23	26.71	24.44

Table 21: Comparison of vectorial dpROF and ROF on Kodak24 using the corresponding BSDS500-derived parameter prescriptions without dataset-specific retuning.

Method	Metric	Noise variance σ^2							
		0.001	0.002	0.004	0.007	0.01	0.02	0.04	0.07
dpROF	SSIM $\uparrow$	0.9207	0.8907	0.8472	0.8167	0.7911	0.7385	0.6799	0.6339
	LPIPS $\downarrow$	0.0801	0.1081	0.1540	0.2122	0.2446	0.3207	0.4001	0.4626
	PSNR $\uparrow$	34.61	32.76	30.88	29.64	28.78	27.16	25.39	23.82
ROF	SSIM $\uparrow$	0.9186	0.8862	0.8439	0.8012	0.7698	0.7271	0.6685	0.6228
	LPIPS $\downarrow$	0.0799	0.1128	0.1610	0.2137	0.2535	0.3306	0.4103	0.4741
	PSNR $\uparrow$	34.44	32.62	30.86	29.46	28.58	26.97	25.23	23.62

Although the absolute metric values vary across datasets, also for the vectorial dpROF model the overall behavior of the model is preserved over the considered noise levels. This suggests that the selected parameters are not strongly tied to the specific image statistics of BSDS500 and show good cross-dataset generalization within the considered natural-image benchmarks. To further assess the cross-dataset behavior of the model, we compare dpROF with the classical ROF model, using in both cases the corresponding parameter prescriptions derived from BSDS500 and without any dataset-specific retuning. On Kodak24, dpROF achieves higher SSIM and PSNR than ROF at 8/8 considered noise levels, while lower LPIPS values are obtained at 7/8 noise levels. On Urban100, the improvement in all metrics

Table 22: Comparison of vectorial dpROF and ROF on Urban100 using the corresponding BSDS500-derived parameter prescriptions without dataset-specific retuning.

Method	Metric	Noise variance σ^2							
		0.001	0.002	0.004	0.007	0.01	0.02	0.04	0.07
dpROF	SSIM $\uparrow$	0.9426	0.9165	0.8791	0.8442	0.8181	0.7563	0.6838	0.6148
	LPIPS $\downarrow$	0.0261	0.0416	0.0689	0.1049	0.1296	0.1947	0.2750	0.3518
	PSNR $\uparrow$	33.26	31.13	29.03	27.35	26.38	24.48	22.56	20.93
ROF	SSIM $\uparrow$	0.9406	0.9126	0.8748	0.8357	0.8064	0.7387	0.6611	0.5879
	LPIPS $\downarrow$	0.0290	0.0473	0.0762	0.1102	0.1377	0.2084	0.2887	0.3650
	PSNR $\uparrow$	33.01	30.88	28.84	27.25	26.26	24.09	22.18	20.52

Table 23: Comparison of vectorial dpROF and ROF on EuroSAT using the corresponding BSDS500-derived parameter prescriptions without dataset-specific retuning.

Method	Metric	Noise variance σ^2							
		0.001	0.002	0.004	0.007	0.01	0.02	0.04	0.07
dpROF	SSIM $\uparrow$	0.9313	0.9022	0.8555	0.8407	0.8145	0.7737	0.7222	0.6837
	LPIPS $\downarrow$	0.0438	0.0565	0.0803	0.1134	0.1305	0.1783	0.2282	0.2693
	PSNR $\uparrow$	36.95	35.11	32.99	32.57	31.59	30.26	28.23	26.43
ROF	SSIM $\uparrow$	0.9275	0.8960	0.8529	0.8083	0.7759	0.7643	0.7155	0.6768
	LPIPS $\downarrow$	0.0438	0.0623	0.0911	0.1247	0.1511	0.1880	0.2349	0.2749
	PSNR $\uparrow$	36.63	34.80	32.97	31.50	30.57	30.08	28.26	26.45

can be seen across the entire noise range. Finally, on EuroSAT dpROF achieves higher SSIM and lower LPIPS values at 8/8 considered noise levels, while higher PSNR values are obtained at 6/8 noise levels.

Finally, we verify how the dpROF model performs on the same pictures across different image resolutions. To this end, we again used the Set14 dataset in two sizes: the originals and the 4x upscaled versions available here [3]. We see that the parameter prescriptions derived from BSDS500, without any dataset-specific retuning, perform well for both low and high resolutions: for the low resolution images, dpROF achieves higher SSIM and PSNR than ROF at 8/8 considered noise levels, while lower LPIPS values are obtained at 4/8 noise levels. For the high resolution images, dpROF outperforms ROF in SSIM at 7/8 noise levels, LPIPS at 2/8 noise levels, and PSNR at 5/8 noise levels. These results show that the performance of the model observed on BSDS500 is largely preserved when the parameter prescriptions are transferred to unseen natural-image datasets and different image resolutions, especially with respect to SSIM and PSNR.

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Table 24: Comparison of vectorial dpROF and ROF on Set14 (low resolution) using the corresponding BSDS500-derived parameter prescriptions without dataset-specific retuning.

Method	Metric	Noise variance σ^2							
		0.001	0.002	0.004	0.007	0.01	0.02	0.04	0.07
dpROF	SSIM $\uparrow$	0.9220	0.8912	0.8485	0.8139	0.7871	0.7281	0.6644	0.6102
	LPIPS $\downarrow$	0.0610	0.0825	0.1155	0.1627	0.1890	0.2545	0.3269	0.3891
	PSNR $\uparrow$	33.98	32.05	30.11	28.76	27.85	26.07	24.19	22.52
ROF	SSIM $\uparrow$	0.9206	0.8892	0.8484	0.8079	0.7785	0.7214	0.6576	0.6023
	LPIPS $\downarrow$	0.0637	0.0880	0.1211	0.1575	0.1859	0.2577	0.3253	0.3852
	PSNR $\uparrow$	33.85	31.97	30.15	28.70	27.78	25.95	24.06	22.31

Table 25: Comparison of vectorial dpROF and ROF on Set14 (high resolution) using the corresponding BSDS500-derived parameter prescriptions without dataset-specific retuning.

Method	Metric	Noise variance σ^2							
		0.001	0.002	0.004	0.007	0.01	0.02	0.04	0.07
dpROF	SSIM $\uparrow$	0.9448	0.9196	0.8800	0.8784	0.8570	0.8281	0.7870	0.7575
	LPIPS $\downarrow$	0.0645	0.1019	0.1715	0.1726	0.2143	0.2553	0.3222	0.3607
	PSNR $\uparrow$	37.98	35.84	33.55	32.52	31.36	29.23	26.71	24.44
ROF	SSIM $\uparrow$	0.9433	0.9175	0.8822	0.8456	0.8185	0.8238	0.7845	0.7544
	LPIPS $\downarrow$	0.0639	0.1004	0.1531	0.2074	0.2458	0.2480	0.3053	0.3435
	PSNR $\uparrow$	37.99	35.91	33.79	32.04	30.90	29.22	26.66	24.30

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